BUBT | BANGLADESH UNIVERSITY OF BUSINESS AND TECHNOLOGY

Committed to Academic Excellence

System Analysis & Design Lab

CSE-318

PROJECT PROPOSAL

Presented for :

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Project Proposal

Project Name: Retail management System

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Project Title: Retail Management System

#INTRODUCTION:

In today's fast-moving retail sector, manual operations create inefficiencies that limit business growth. This project proposes a digital Retail Management System to automate daily store activities such as inventory tracking, customer billing, and sales reporting. Built with a modern tech stack, this system will offer a smooth and responsive user experience while ensuring data security and performance through a structured backend. It will serve as a centralized platform for store owners to manage sales, inventory, and customers in real time. The system aims to be intuitive, scalable, and tailored to meet the needs of small and medium-sized businesses.

#PROBLEM STATEMENT:

Retail shops often face the following challenges:

- Manual errors in inventory updates and sales records. leading to stock mismatches and financial discrepancies.
- Time-consuming billing processes that reduce customer satisfaction and increase service delays.
- Inability to generate accurate and timely sales or stock reports, affecting business decisions.
- Outdated data systems that lead to poor customer service and inefficient workflow.
- Lack of centralized data access, making it hard to monitor operations across departments or branches.
- Difficulty in maintaining and tracking customer information and purchase history.

The proposed Retail Management System aims to solve these problems by providing a digital platform with automation, real-time updates, and secure, centralized data management.

#Existing Systems Analysis

Existing System	Key Features	Gaps / Limitations
Shopify	E-commerce focus, inventory tools, third-party plugins	High cost, not ideal for local offline retail
Tally ERP	Inventory & accounting	Complex UI, not cloud/web-based
Square POS	Billing, sales analysis, mobile friendly	Limited inventory functions, expensive hardware needed

#Identified Limitations:

- Not suitableforsmall/local businesses.
- Lack of customization and user-friendliness.
- Hardware dependency or high costs.

#Key Features:

1.Product & Inventory Management:

- Add, edit, delete product listings.
- Track stock levels automatically after each sale.
- Notify when stock is low.

2.Billing and Sales Module:

- Generate real-time customer bills.
- Calculate totals, taxes, and apply discounts.
- Save and retrieve previous bills.

3.Customer Management:

- Register customers and save their data.
- View purchase history.
- Targeted service based on customer records.

4.Admin Dashboard:

- User authentication and role-based access (admin/staff).
- View system status, track performance.
- Manage staff user accounts.

5. Sales Reporting and Analytics:

- Generate daily/weekly/monthly reports.
- View most sold products, revenue trends.
- Export reports in CSV/PDF

6. Secure Login System:

- Password hashing and session management.
- Prevent unauthorized access.
- Logout and session timeout features.

7. Responsive Web Design:

- Mobile and desktop-friendly layout.
- Simple, intuitive UI for non-technical users.
- Works across browsers and devices.

#Technologies Used:

1. Frontend:

- HTML5: For structuring web pages.
- CSS3: For styling and responsive layout.
- JavaScript: For interactivity and client-side logic.

2. Backend:

- Python (Flask/Django): For handling server-side logic, APIs, authentication, and business rules.

3. Database:

- Mobile and desktop-friendly layout.
- Simple, intuitive UI for non-technical users.
- Works across browsers and devices.

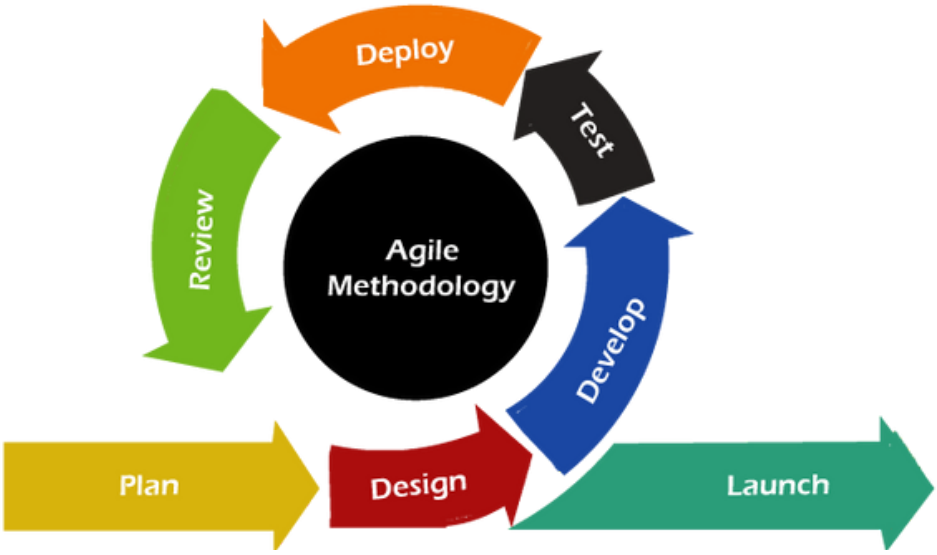
4. Development Approach:

AgileModel:Incrementaland iterative development for better flexibility and feedback.

#Reason of chosing Agile Model:

1. Flexibility for Changes – Agile allows us to adapt quickly to changing requirements.
2. Incremental Delivery – Features are delivered in small, manageable sprints.
3. Better Collaboration – Continuous communication keeps all team members aligned.
4. Early Detection of Errors – Regular testing helps catch bugs early in development.
5. Improved User Satisfaction – Frequent feedback ensures the system meets user needs.
6. Risk Management – Small iterations help identify and manage risks effectively.
7. Time-Efficient – Short cycles reduce rework and speed up overall delivery.

#Agile model diagram:



#Conclusion:

Store operations will be simplified and automated by the **Retail Management System,** which will enable retailers to manage stock, process sales and understand customer behaviour. Due to the webbased ui and the stable backend retaillunch is designed for small to medium businesses. The implementation of the Agile model makes the system to be developed in iterations, this accommodates for the continuous improvement and feedback. The proposed system is low-cost, customizable, and scalable, suitable for the modern day retail requirements.